Experiment 19 :Experiment to understand Logistic Regression for a given data set.

Description

The program demonstrates Logistic Regression for predicting whether a customer will purchase a product based on age and salary. The dataset is divided into training and testing data, and a Logistic Regression model is trained and evaluated.

Program:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report
# New dataset
data = {
    "Age": [19, 21, 24, 27, 29, 31, 34, 37, 39, 41, 44, 47, 49, 53, 56],
    "Salary": [22000, 26000, 30000, 33000, 38000, 42000, 48000, 53000,
               58000, 63000, 68000, 73000, 78000, 87000, 92000],
    "Purchased": [0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1]
}
df = pd.DataFrame(data)
print("Dataset:")
print(df)
print("\nFirst 5 Rows:")
print(df.head())
# Features and label
features = df[["Age", "Salary"]].values
label = df["Purchased"].values
# Split the dataset
x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=42
)
# Create Logistic Regression model
model = LogisticRegression(max_iter=1000)
```

```
# Train the model
model.fit(x_train, y_train)
# Accuracy
train_score = model.score(x_train, y_train)
test_score = model.score(x_test, y_test)
print("\nTraining Accuracy:", train_score)
print("Testing Accuracy:", test_score)
# Classification Report
y_pred = model.predict(x_test)
print("\nClassification Report:")
print(classification_report(y_test, y_pred, zero_division=0))
```

Output:

```
Dataset:
    Age  Salary  Purchased
0    19   22000          0
1    21   26000          0
2    24   30000          0
3    27   33000          0
4    29   38000          0
5    31   42000          0
6    34   48000          1
7    37   53000          1
8    39   58000          1
9    41   63000          1
10   44   68000          1
11   47   73000          1
12   49   78000          1
13   53   87000          1
14   56   92000          1

First 5 Rows:
   Age  Salary  Purchased
0   19   22000          0
1   21   26000          0
2   24   30000          0
3   27   33000          0
4   29   38000          0

Training Accuracy: 1.0
Testing Accuracy: 1.0

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         1
           1       1.00      1.00      1.00         2

    accuracy                           1.00         3
   macro avg       1.00      1.00      1.00         3
weighted avg       1.00      1.00      1.00         3
```